



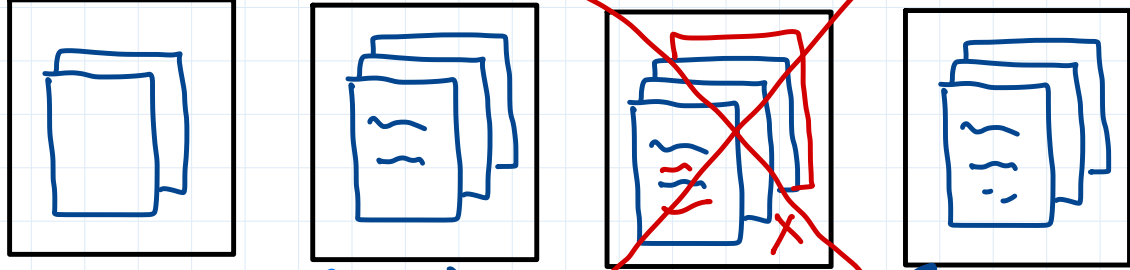
GIT

Trainer: Nilesh Ghule

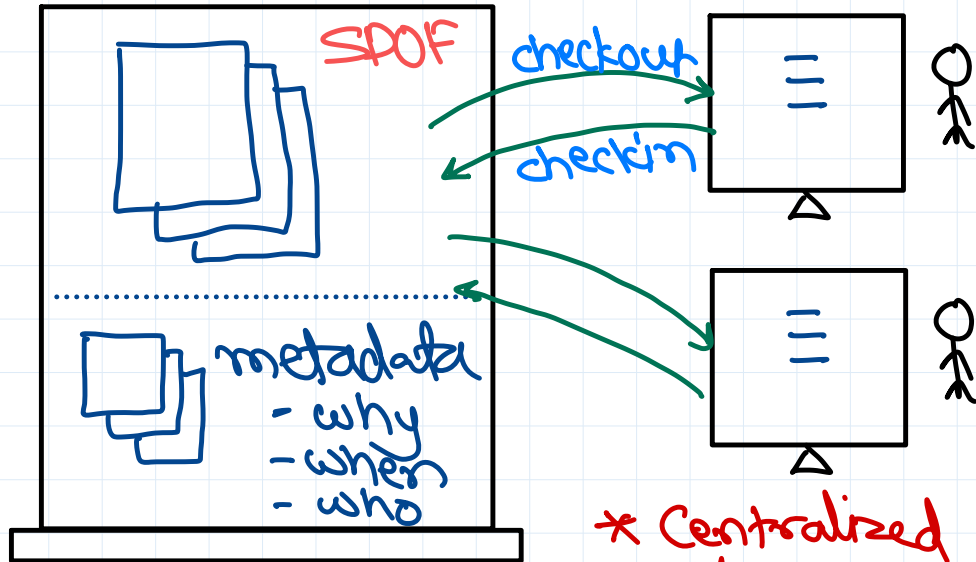


Version Control System

19-aug 20-aug 21-aug 22-aug

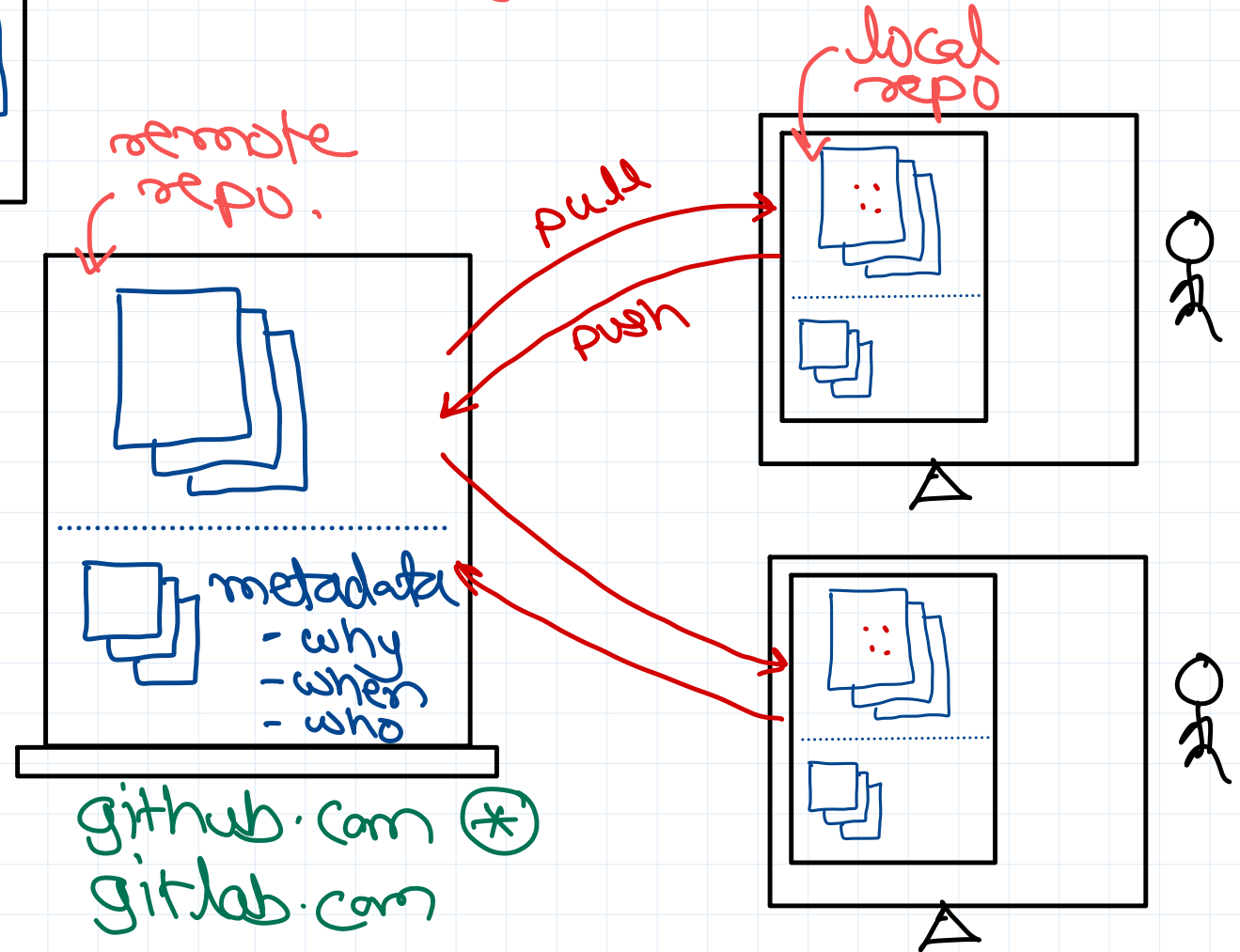


* manual VCS



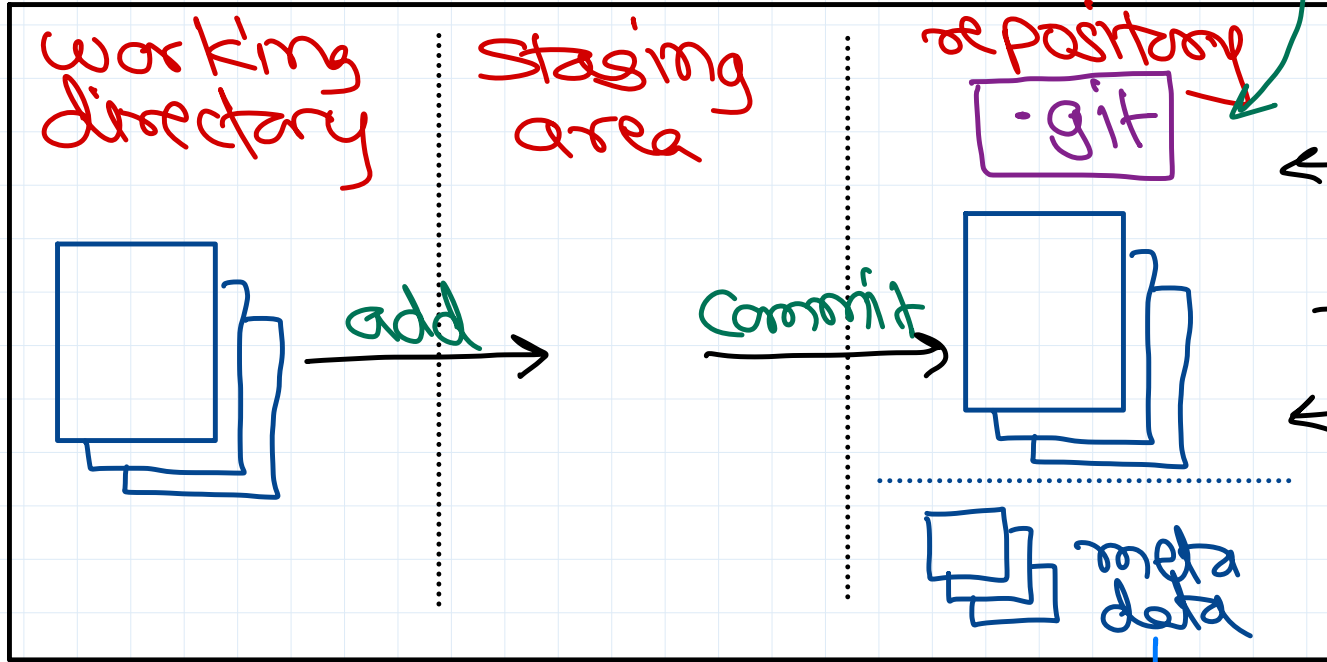
* Centralized VCS

* Distributed VCS
e.g. GIT.



GIT architecture

local repo.



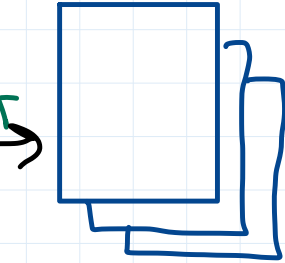
git config
→ user.name
→ user.email

local machine

- ① git bash
 - ② IDE (VScode, Eclipse, ...)
- git init

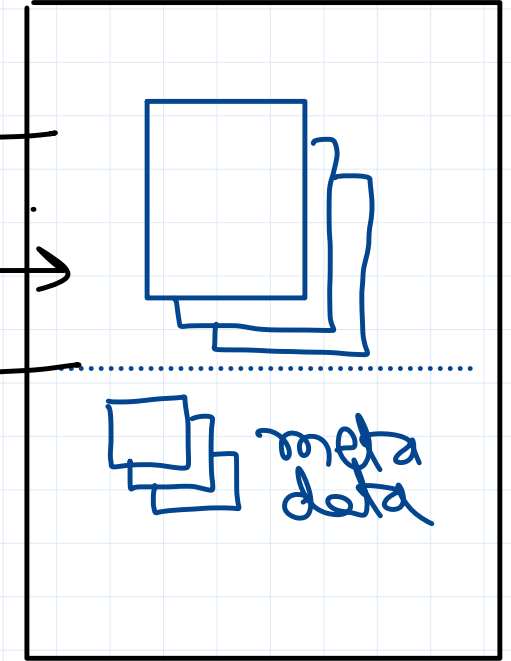
repository

`.git`



- when
- why
- who

remote repo.
(github.com)



clone

push

pull

- git status
- git log

Git installation & setup

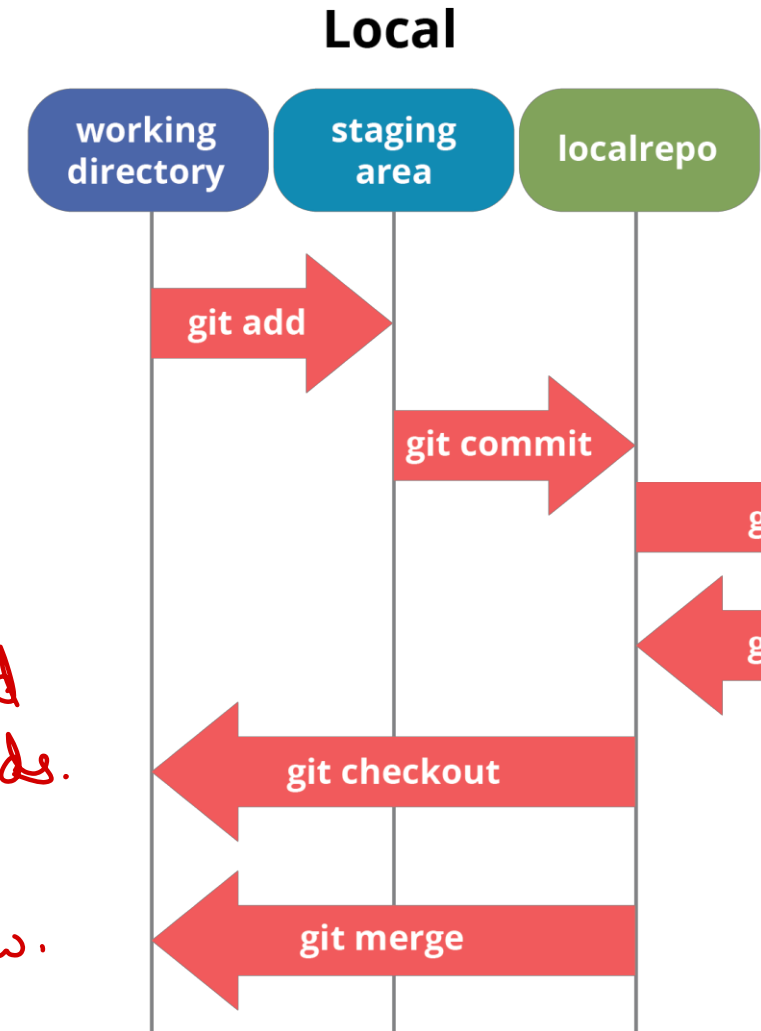
- On Ubuntu
 - `sudo apt-get install git`
 - List global settings
 - `git config --global --list`
 - Set up global properties
 - `git config --global user.name <your name>`
 - `git config --global user.email <your email>`
 - `git config --global core.editor <editor app>`
 - GIT user details are associated with each commit done by the user.
- On Windows
 - Download and install GIT.
 - <https://git-scm.com/downloads>
 - Installed components
 - GIT bash
 - git-gui + gitk
 - GIT Bash
 - git command
 - bash commands
 - vim editor
- In editor/IDE
 - All leading IDEs have GIT support.
 - VS Code, Eclipse, ...



GIT commands

- terminal> git init
 - terminal> git status
 - terminal> git status -s
 - terminal> git add <file-path>
 - terminal> git add <dir-path>
 - terminal> git commit -m "message"
- terminal> git diff (track changes that are not staged)
 - terminal> git checkout <file-path> (discard changes & get last committed version)
 - terminal> git reset (unstage the changes)
 - terminal> git reset --hard (unstage the changes and replace with last committed version)

← advanced
Commands.
ignore
for now.



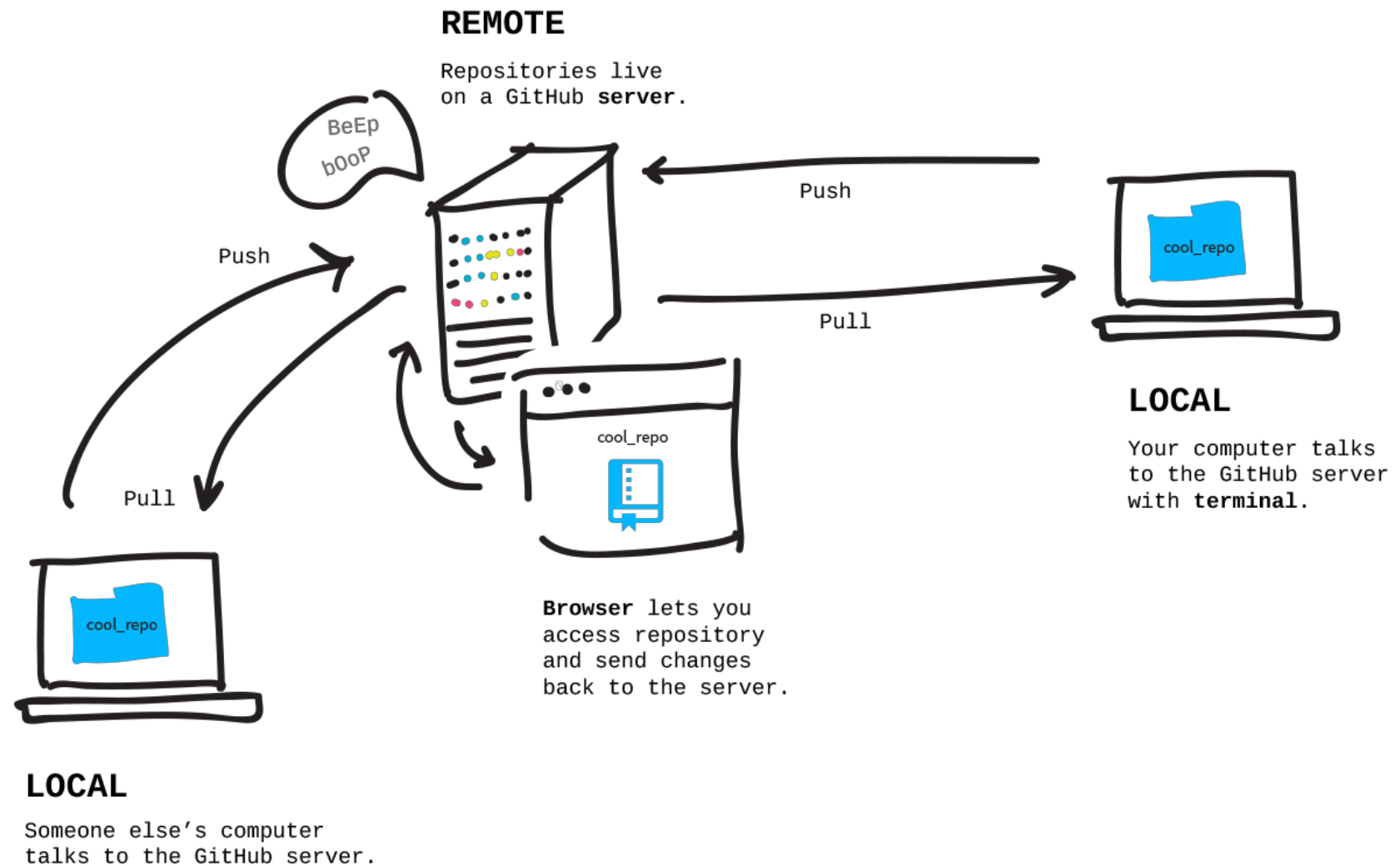
.gitignore

- List directories or files to be ignored for git repository.
- Used to ensure that binaries, IDE metadata files and other undesired files are not maintained in git repository.



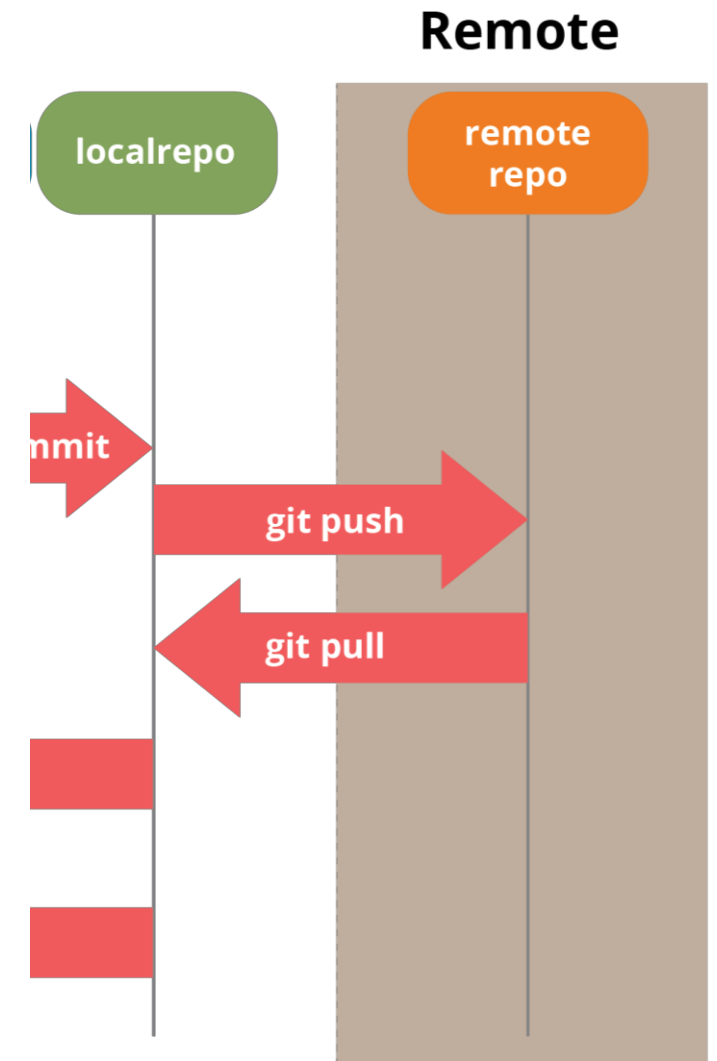
GIT Remote repository

- To maintain code repository at centralized location (for code sharing).
- Can be in intranet or internet.
- Popular vendors
 - github.com
 - gitlab.com
 - bitbucket.org



GIT commands

- `git remote add origin <remote url>`
- `git remote -v`
- `git clone <url>`
- `git push origin <branch>`
- `git push`
- `git pull origin <branch>`
- `git pull`



GIT workflow

- Create project on gitlab.
- Clone repository on local machine.
- Add/modify code locally.
- Commit code in local repository.
- Push code to gitlab repository.
- Other developers can pull your code.





Thank you!

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